THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE UNIT TEST SET C

Chapters 10–13: **Light: Mirrors and Lenses, Keeping Time with the Skies, How Nature Works in Harmony, Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. Which can form a real image on a screen? 1

(a) concave lens **(b)** plane mirror **(c)** convex mirror **(d)** convex lens

2. A lens is a piece of transparent material with: 1

(a) curved surface(s) **(b)** a rough surface only
(c) no surface **(d)** a flat surface only

3. Corrective glasses for nearsighted eyes use a: 1

(a) concave lens **(b)** plane mirror **(c)** convex lens **(d)** convex mirror

4. The phase when the Moon is not visible is the: 1

(a) new moon **(b)** gibbous **(c)** crescent **(d)** full moon

5. The ‘Father of the Indian Space programme’ is: 1

(a) S. Chandrasekhar **(b)** Homi Bhabha **(c)** Vikram Sarabhai **(d)** C.V. Raman

6. The Indian National Calendar was adopted from: 1

(a) 26 January 1950 **(b)** 1 January 1947 **(c)** 15 August 1947 **(d)** 21 March 1956

7. Invasive species harm ecosystems because they: 1

(a) fix nitrogen **(b)** increase biodiversity
(c) purify water **(d)** compete with native species

8. National parks and wildlife sanctuaries are examples of: 1

(a) pollution **(b)** deforestation
(c) invasive species **(d)** conservation of ecosystems

9. First trophic level is occupied by: 1

(a) decomposers **(b)** herbivores **(c)** carnivores **(d)** producers

10. Greenhouse gases mentioned include carbon dioxide and: 1

(a) methane **(b)** helium **(c)** neon **(d)** argon

11. A small rise in global temperature can: 1

(a) remove gravity **(b)** disturb climate patterns
(c) stop all seasons forever **(d)** increase ozone

12. Earth supports life mainly because of its: 1

(a) distance from the Sun, atmosphere and water
(b) number of moons
(c) size of oceans only
(d) colour from space

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. What is lateral inversion? 2

14. Differentiate a lunar and a solar calendar. 2

15. Define mutualism with an example. 2

16. What is the habitable zone? 2

17. Why is Earth's gravity important for life? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Describe the activity to verify the laws of reflection. 3

19. Why do festivals like Eid-ul-fitr occur on different Gregorian dates each year? 3

20. Compare mutualism, commensalism and parasitism in a table with one example each. 3

21. What is regeneration? Give an example. 3

22. Explain why the Moon's bright part shrinks after purnima. 3

OR

Name three Indian satellites and their purpose. 3

23. Write three practices of sustainable farming. 3

OR

Describe the problems of using pesticides and monoculture. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. Why do some festivals fall on nearly the same Gregorian date each year while others move? Give examples. 5

OR

Describe how lunar, solar and luni-solar calendars work. Give one feature of each. 5

25. Describe the greenhouse effect, its natural role and how human activities enhance it. 5

OR

Compare reproduction in fish, frogs, birds and mammals on fertilisation and embryo development. 5

END OF QUESTION PAPER Total: 50 Marks

MASTERMIND